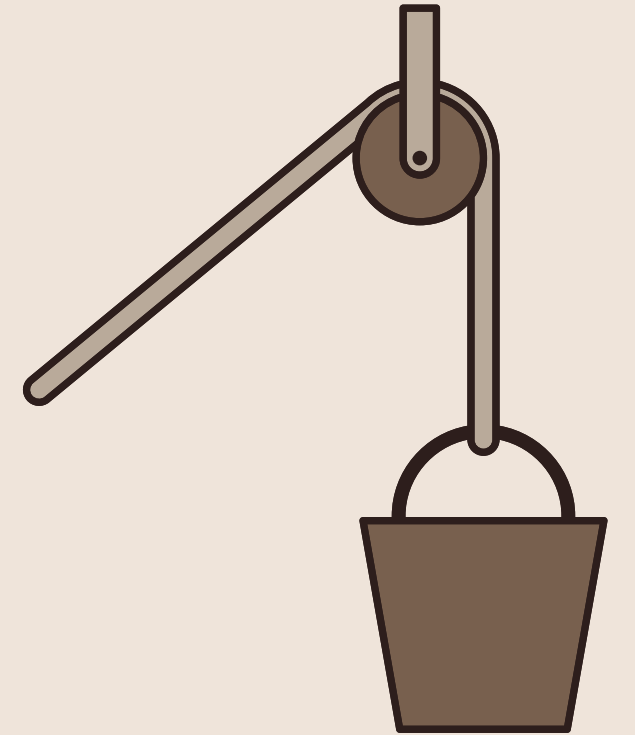
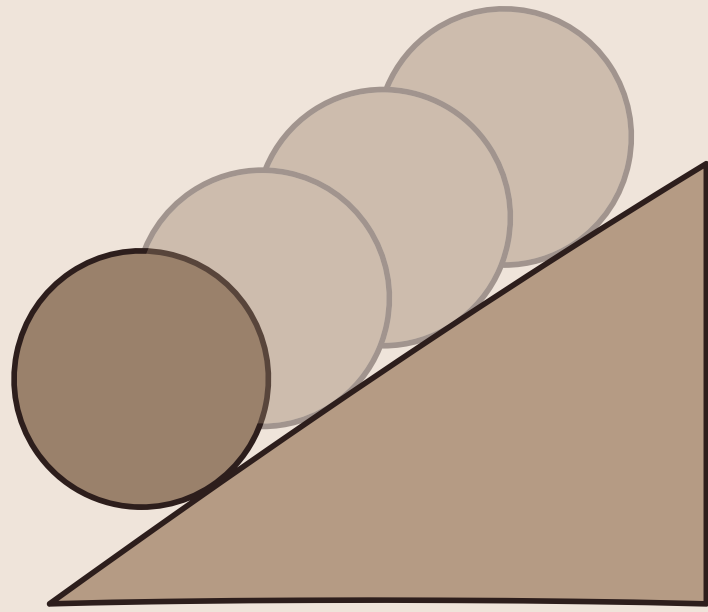
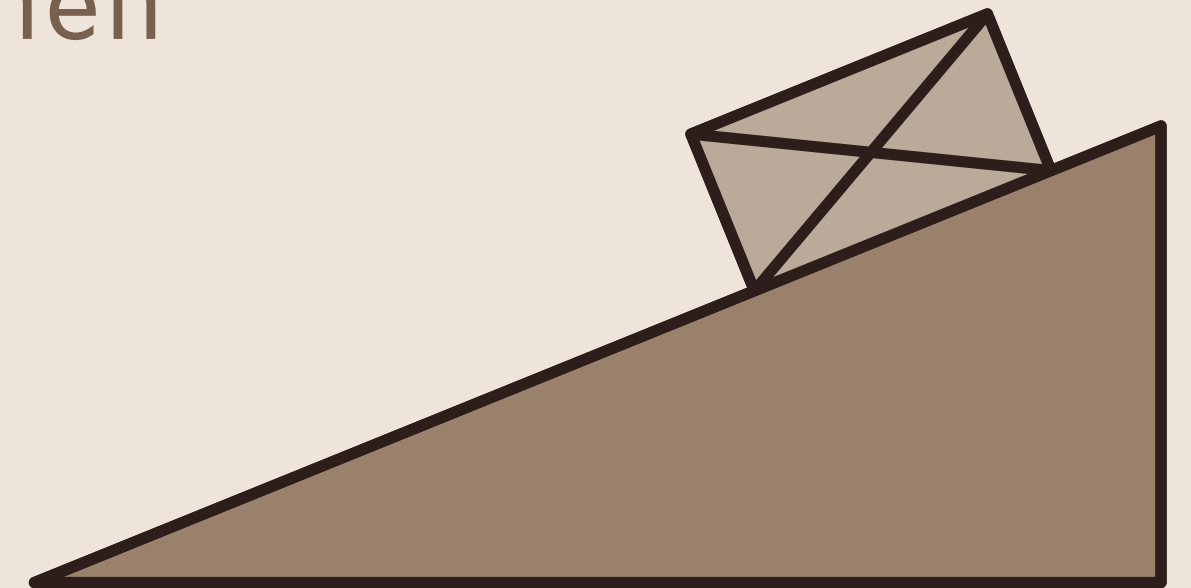
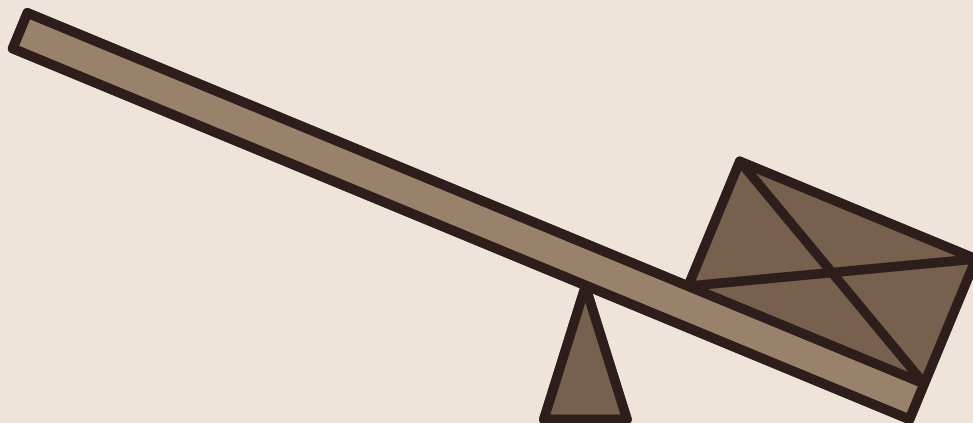
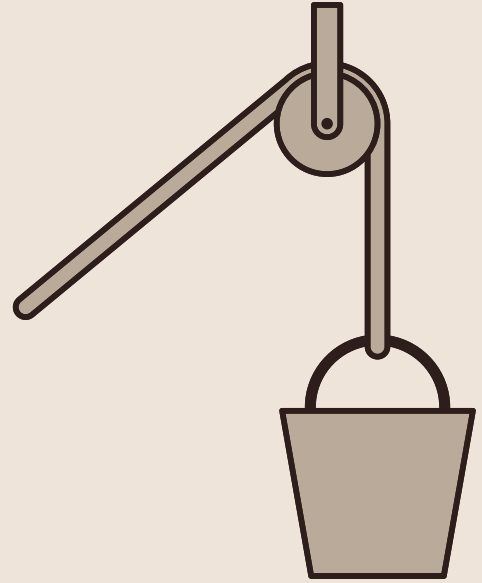




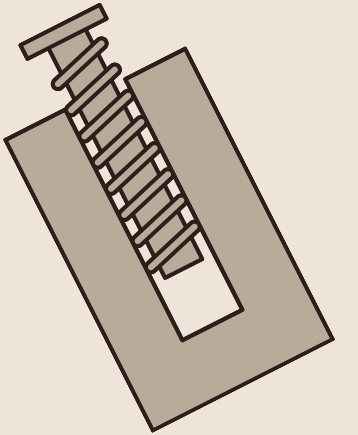
PREDICTION OF TENSILE STRENGTH OF SAWDUST ASH CONCRETE USING MACHINE LEARNING (KNIME)

Presented by Adebayo Stephen
Adedolapo

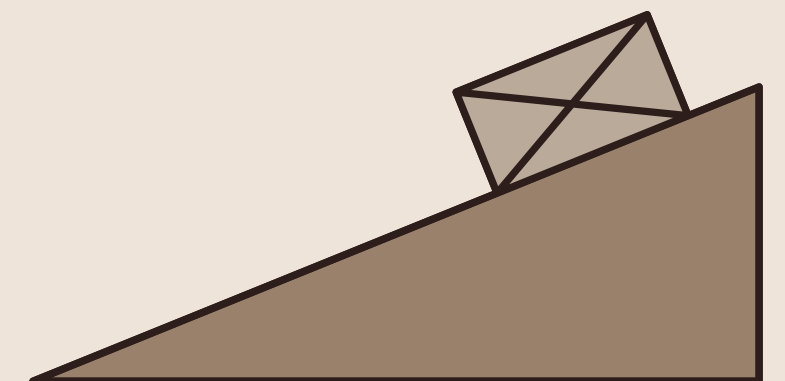
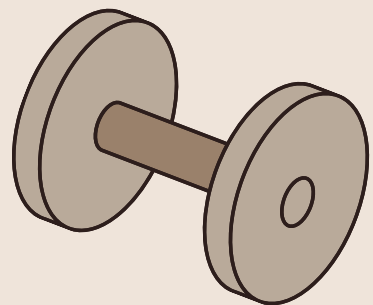


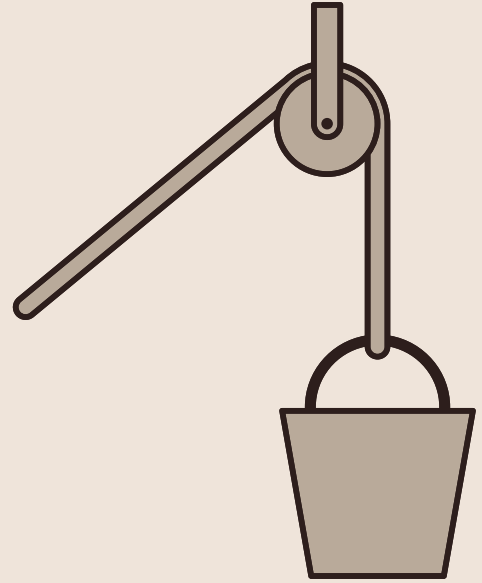


PROBLEM STATEMENT

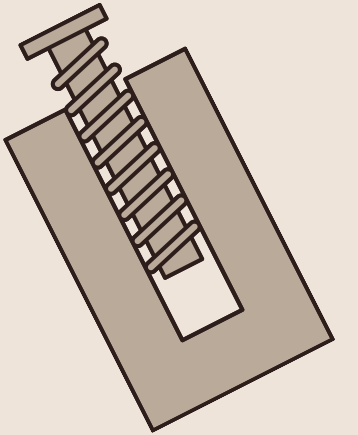


- Need for sustainable concrete materials
- Sawdust ash as partial cement replacement
- Difficulty of predicting tensile strength experimentally

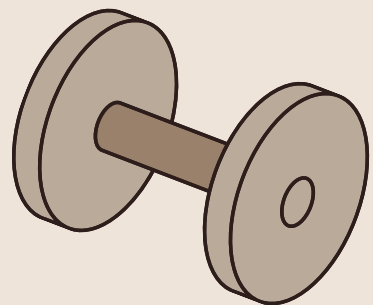
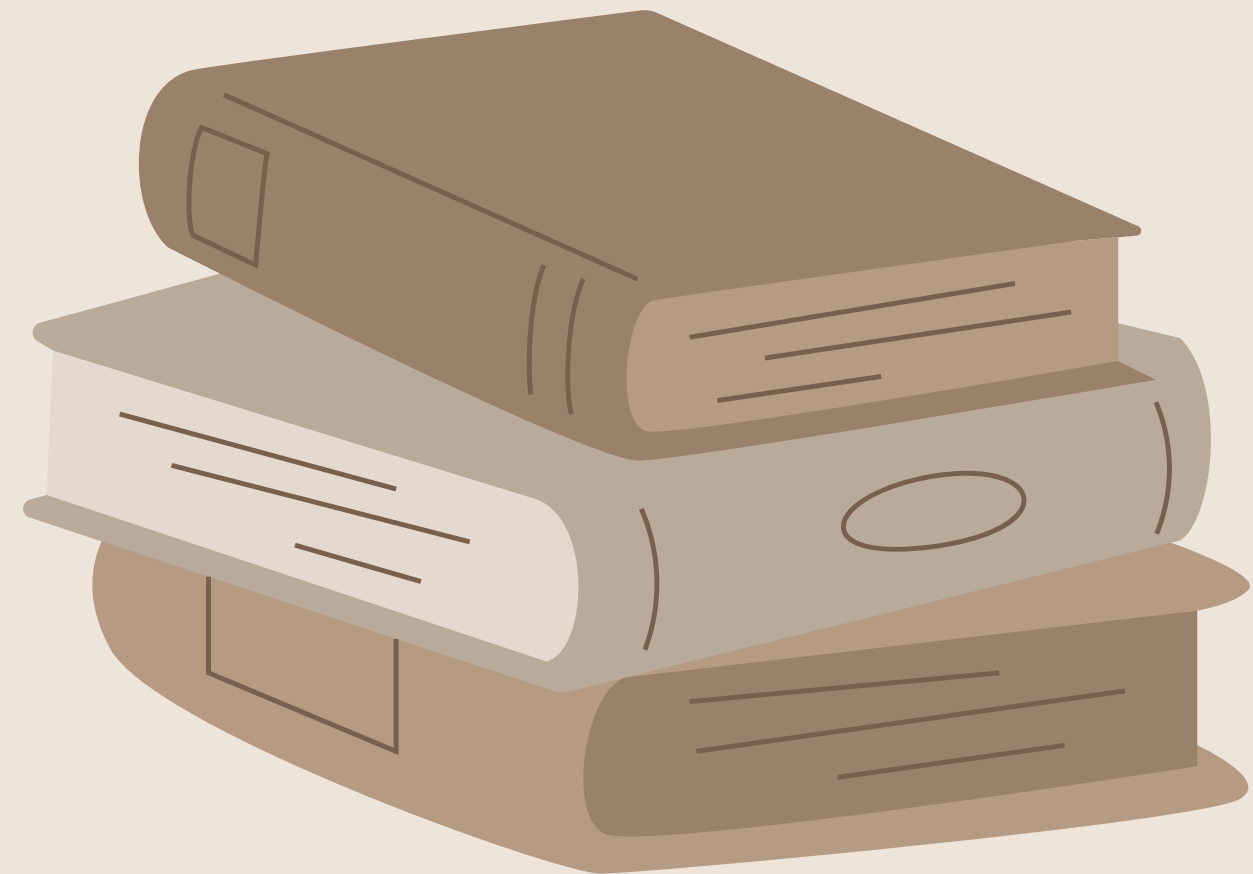


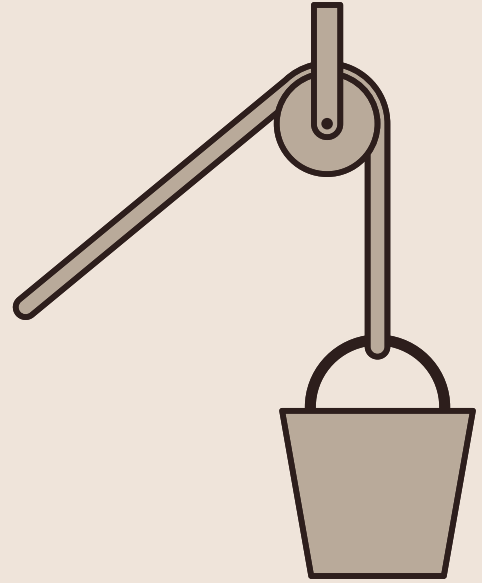


OBJECTIVES

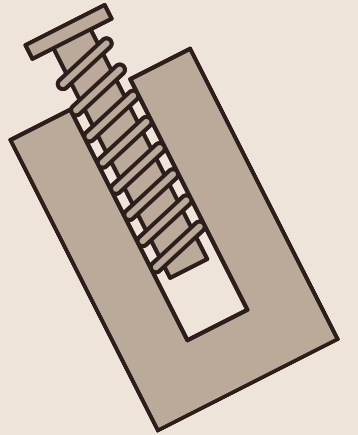


- Predict tensile strength using ML
- Evaluate effect of sawdust ash
- Compare regression models





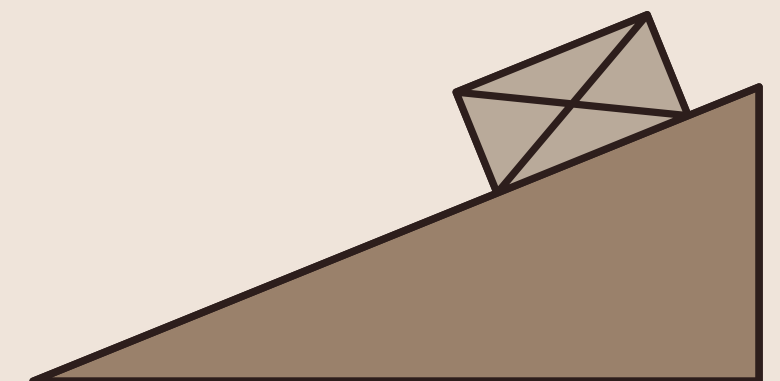
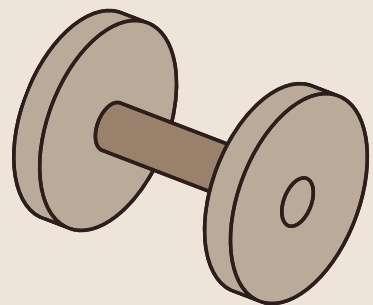
DATASET



The dataset used in this study is adapted from the UCI Concrete Strength Dataset, which is publicly available and widely used in civil engineering and machine learning research. The dataset contains records of concrete mixtures with varying proportions of cement, blast furnace slag, supplementary cementitious material, water, superplasticizer, coarse aggregate, fine aggregate, and curing age. These input variables are used to predict the mechanical strength of concrete.

Due to the limited availability of publicly accessible datasets specifically containing sawdust ash concrete properties, the fly ash component in the original dataset was adapted to represent sawdust ash as a supplementary cementitious material. Additionally, the dataset was repurposed to predict tensile strength instead of compressive strength for modeling purposes. This adaptation provides a realistic baseline for evaluating the feasibility of machine learning models in predicting the tensile strength of sustainable concrete incorporating sawdust ash.

Source to original dataset before adaption:
<https://www.kaggle.com/datasets/vivekgediya/concrete-data>



METHODOLOGY (DIAGRAM)

THIS STUDY UTILIZES A STRUCTURED MACHINE LEARNING WORKFLOW IN KNIME TO PREDICT THE TENSILE STRENGTH OF SUSTAINABLE CONCRETE CONTAINING SAWDUST ASH. THE PROCESS BEGAN WITH DATA INGESTION AND STANDARDIZATION USING A DOMAIN CALCULATOR, FOLLOWED BY NORMALIZATION TO ELIMINATE SCALE BIAS AND PARTITIONING THE DATA INTO TRAINING AND TESTING SETS. RANDOM FOREST REGRESSION WAS IMPLEMENTED AS THE CORE ALGORITHM, LEVERAGING AN ENSEMBLE OF DECISION TREES TO ACCURATELY MAP THE COMPLEX, NONLINEAR RELATIONSHIPS BETWEEN MIX CONSTITUENTS AND MECHANICAL STRENGTH. AFTER GENERATING PREDICTIONS, THE MODEL'S PERFORMANCE WAS VISUALLY VALIDATED THROUGH SCATTER PLOTS COMPARING PREDICTED VERSUS ACTUAL VALUES, WITH FINAL RESULTS EXPORTED VIA A CSV WRITER TO SUPPORT DATA-DRIVEN DECISION-MAKING IN CIVIL ENGINEERING.

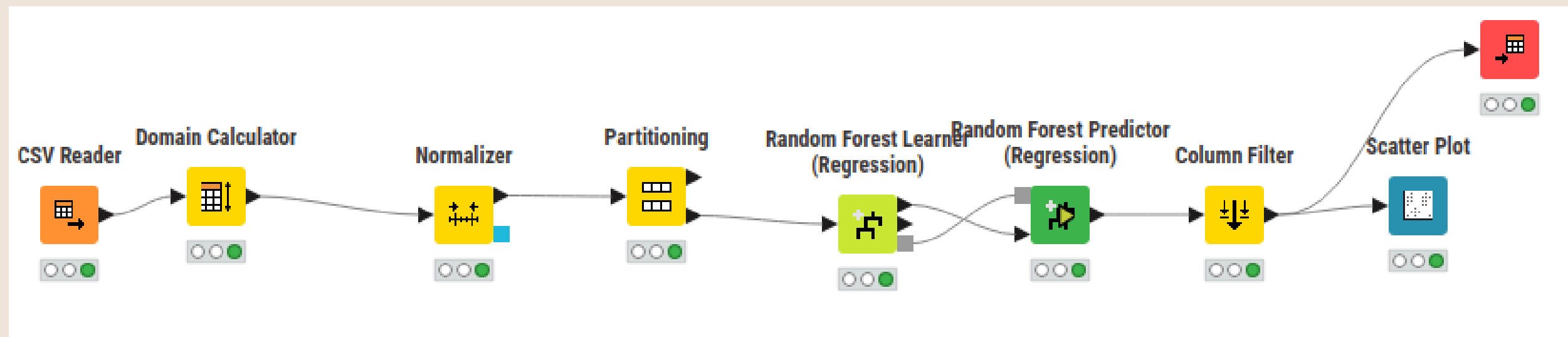
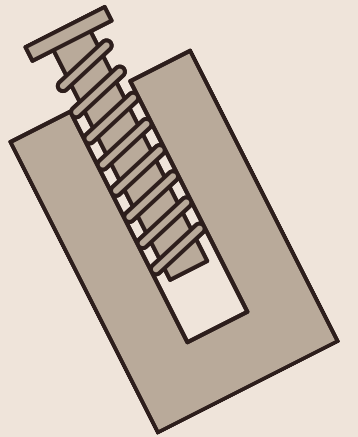
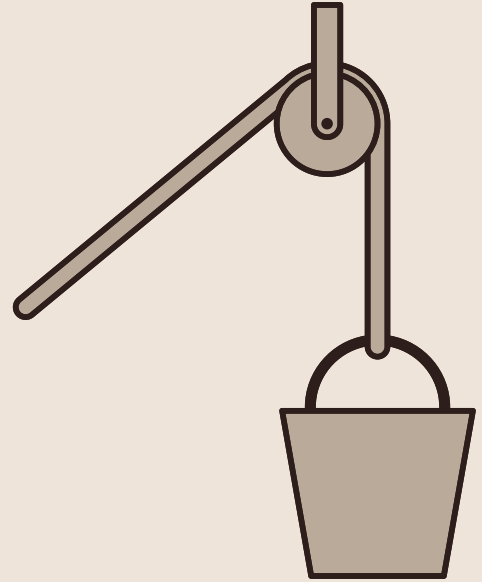


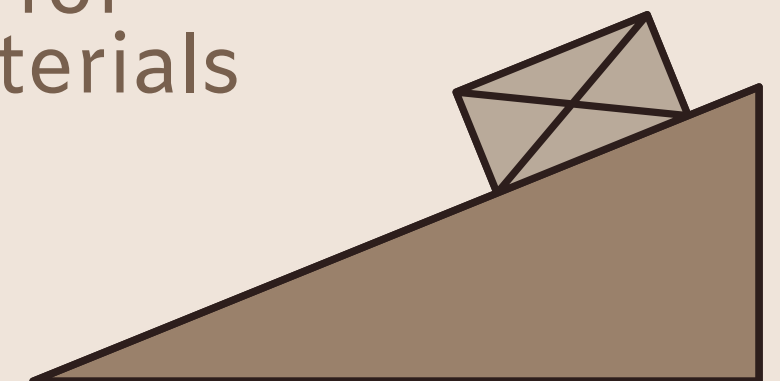
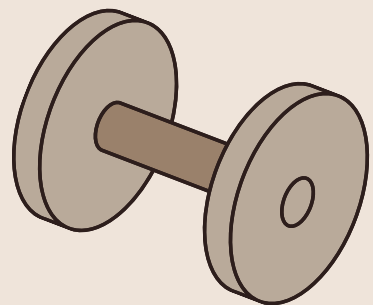
Image Credit: My Knime Prediction

MACHINE LEARNING MODEL

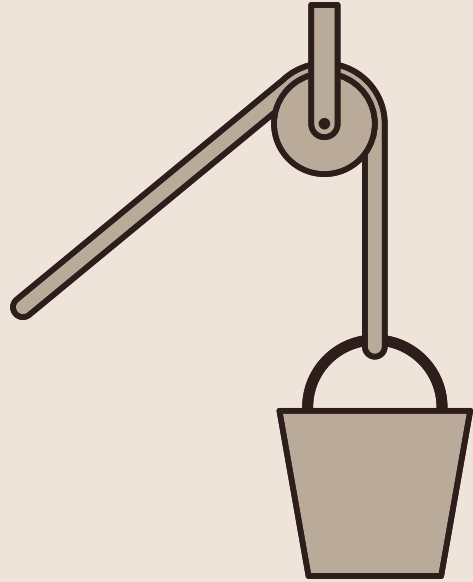


Random Forest Regression

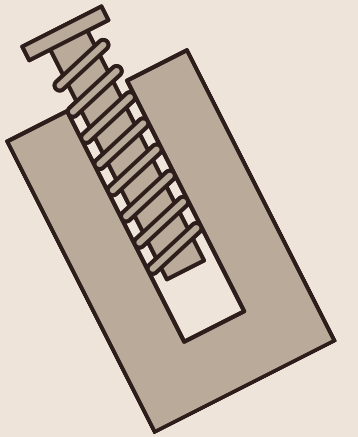
Random Forest Regression is an ensemble machine learning model that combines multiple decision trees to improve prediction accuracy and robustness. It captures complex and non-linear relationships between concrete mix components and tensile strength by averaging predictions from many trees trained on different subsets of the data. In this project, Random Forest regression is used to achieve higher predictive accuracy and to evaluate feature importance, making it particularly suitable for modeling the complex behavior of sustainable concrete materials containing sawdust ash.



RESULTS

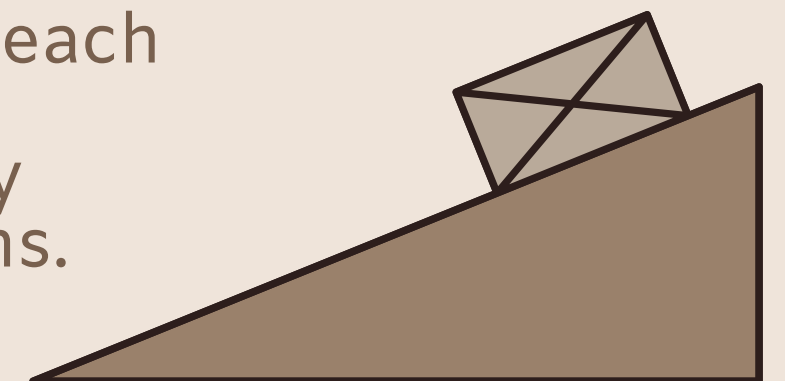
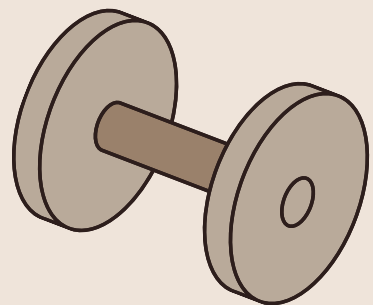


The results obtained from the Random Forest Regression model demonstrate exceptional accuracy in predicting the tensile strength of sustainable concrete incorporating sawdust ash. By analyzing the output data, it is evident that the model's predictions closely align with the actual laboratory measurements, frequently achieving a precision margin within 0.2 to 0.5 MPa. This high level of accuracy confirms that the Random Forest algorithm successfully captured the complex, nonlinear interactions between material components—specifically the relationship between traditional cement, curing age, and the supplementary sawdust ash. Ultimately, these results provide a robust proof of concept, showing that machine learning can reliably simulate physical laboratory tests, allowing for the rapid optimization of eco-friendly concrete mixes while significantly reducing experimental costs and the traditional 28-day waiting period.



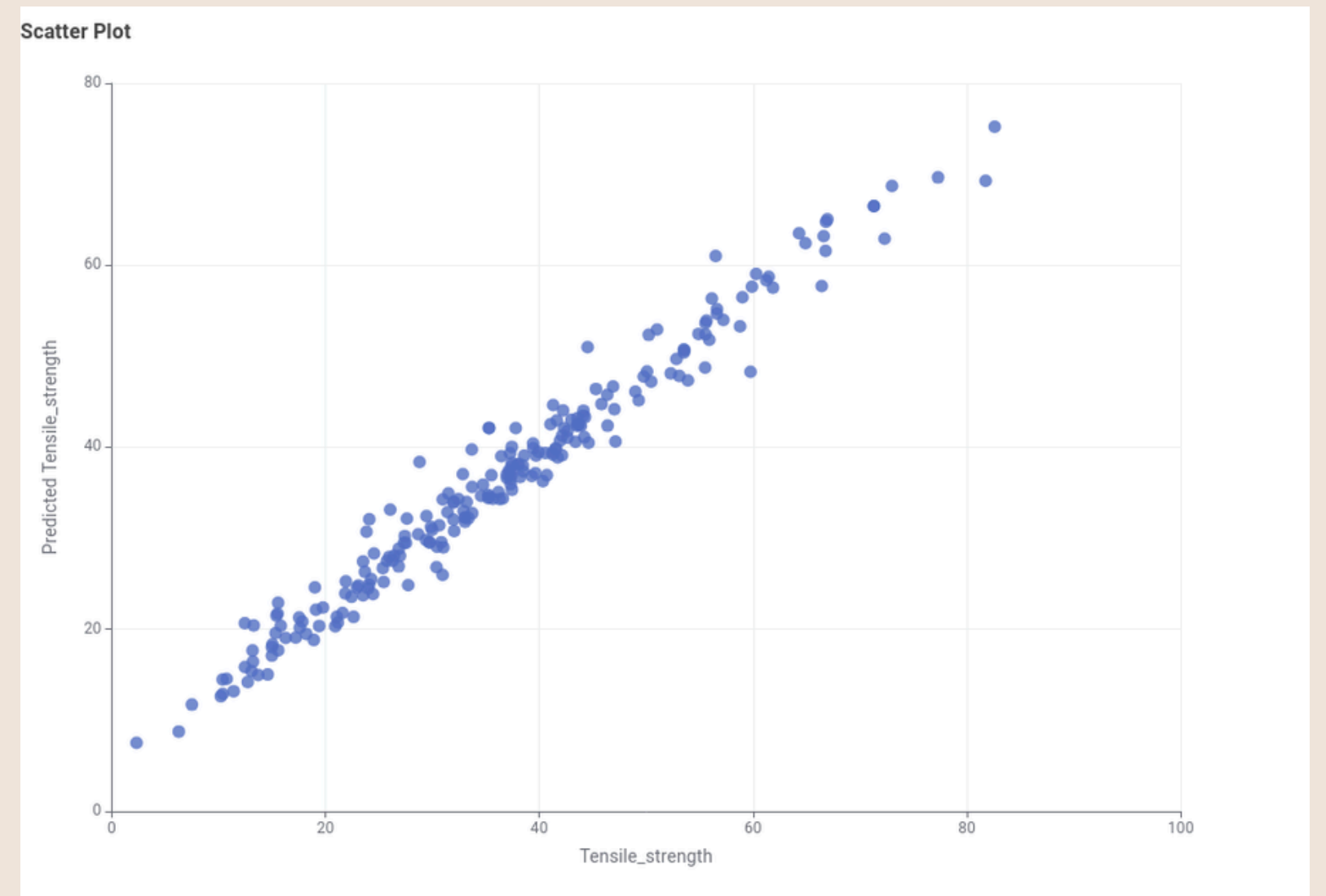
Why Random Forest is your best result:

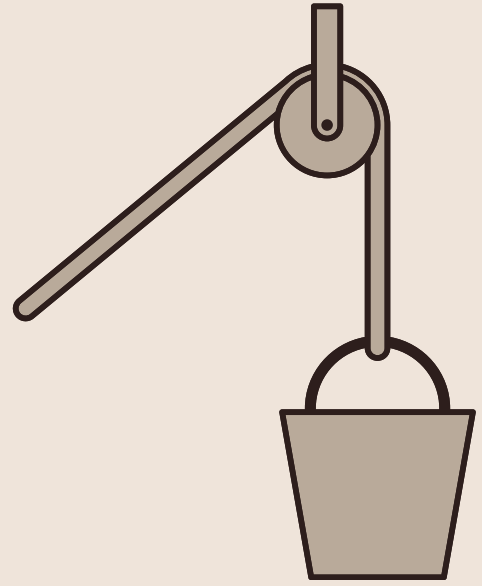
1. **Nonlinearity:** Concrete strength doesn't increase in a simple straight line; it depends on complex chemical reactions. Random Forest "learns" these patterns better than simple linear models.
2. **Robustness:** It uses 100 different decision trees (an ensemble) to reach a consensus, which is why your error margin is so low.
3. **Engineered for Materials:** It is less sensitive to "noise" in laboratory data, making it the superior choice for material science applications.



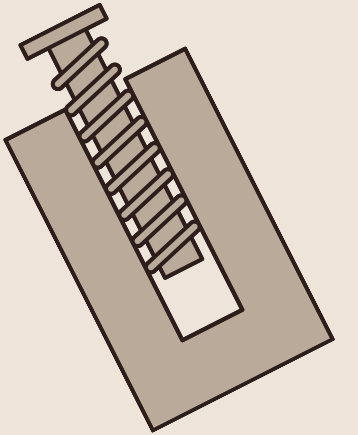
FEATURE IMPORTANCE

The scatter plot provides a visual validation of the Random Forest model's performance by plotting actual laboratory results against predicted tensile strength. The tight clustering of data points along the 45-degree identity line confirms the model's high accuracy and its ability to generalize patterns across diverse concrete mixes. This strong correlation demonstrates that the model successfully captured the complex contributions of cement, curing age, and sawdust ash to the material's strength. Ultimately, this visualization proves that the machine learning model can reliably simulate physical strength tests, offering a more efficient and sustainable approach to material optimization.

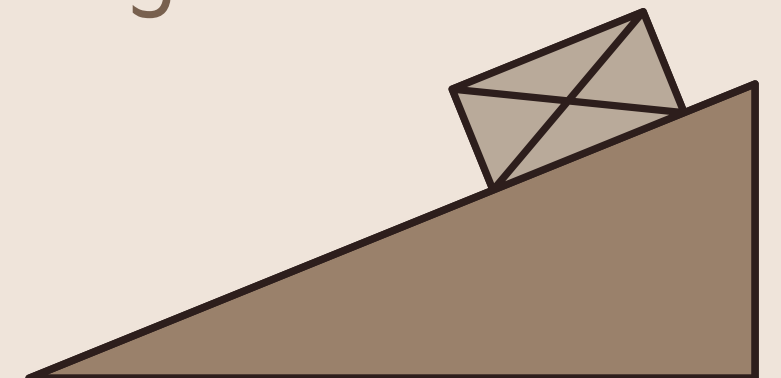
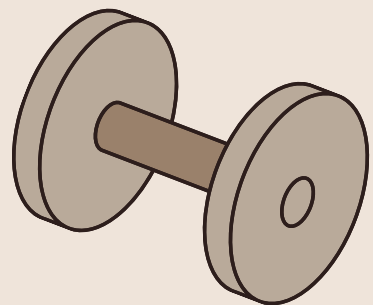


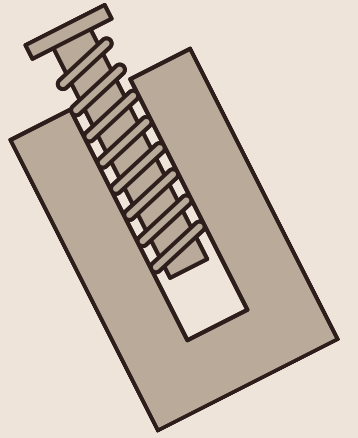
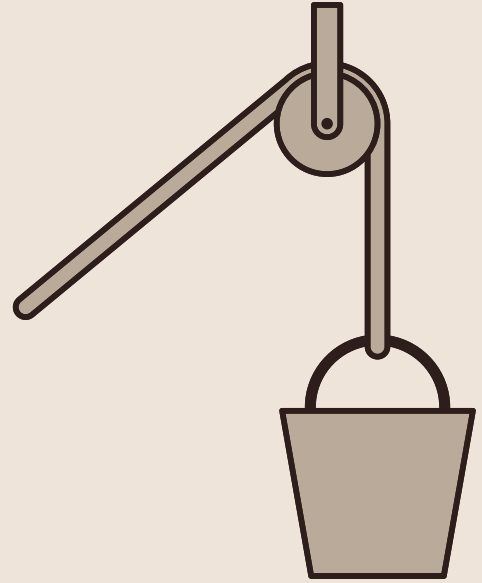


KEY LEARNINGS



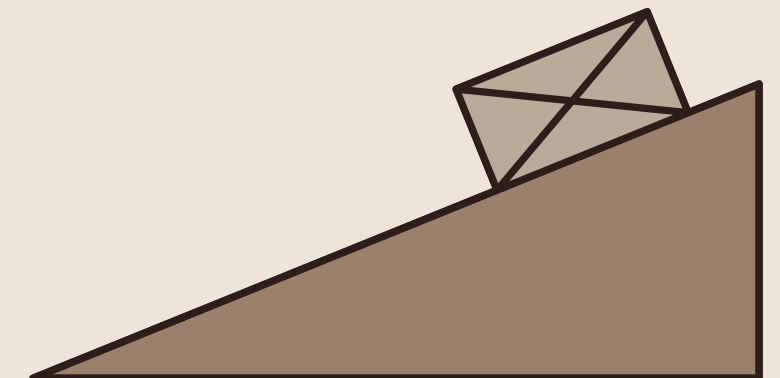
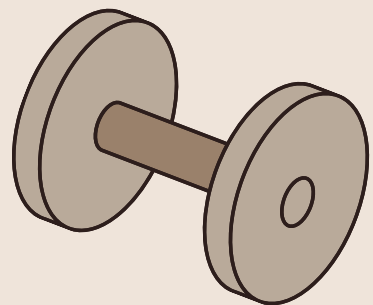
This project demonstrated that machine learning can effectively model the complex and non-linear behavior of concrete materials. Random Forest regression outperformed Linear Regression, highlighting the importance of using advanced models for material property prediction. The study also showed that sawdust ash can influence tensile strength alongside traditional concrete components, supporting its potential use in sustainable construction. Overall, the project improved understanding of data preprocessing, model evaluation, and the application of ML tools such as KNIME in civil engineering research

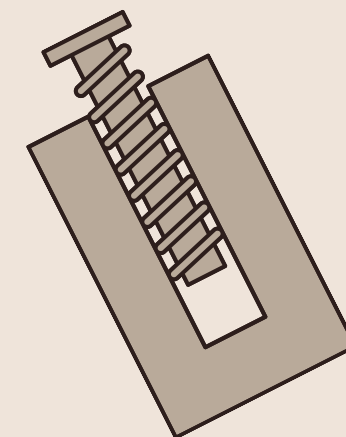
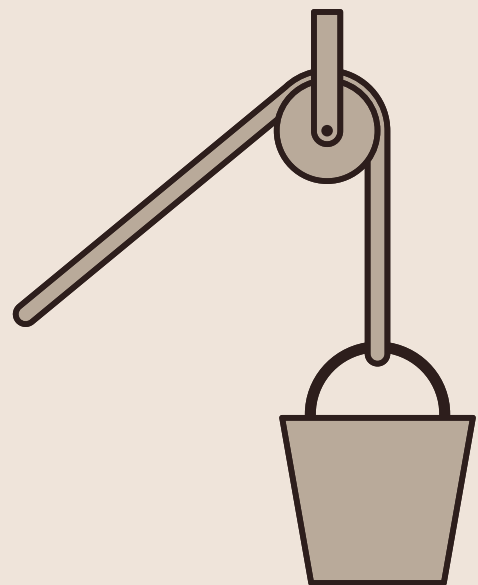




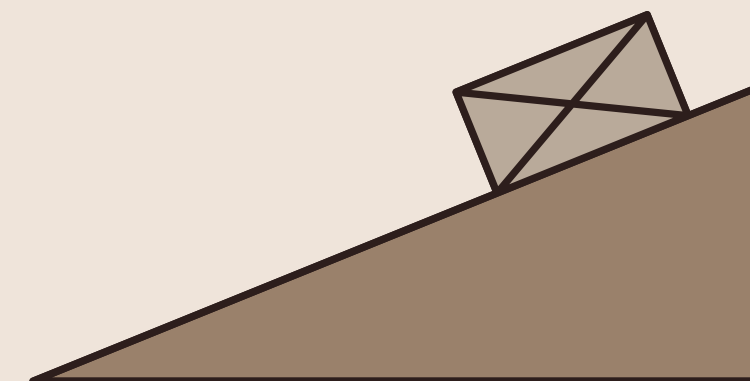
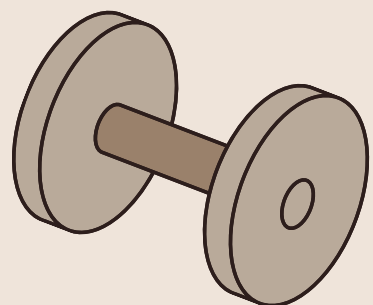
CONCLUSION

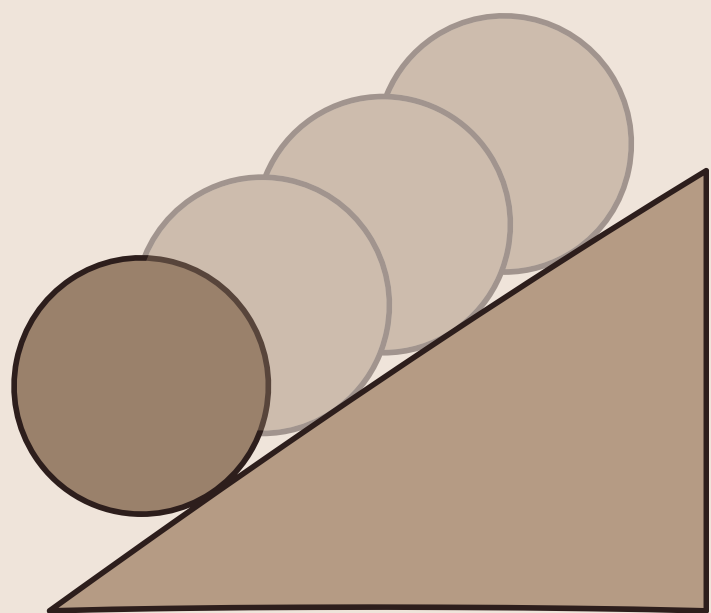
- ML successfully predicts tensile strength
- Sawdust ash shows potential





QNA SESSION





THANK
YOU

